PSYC 6273 – Computer programming for experimental psychology
Fall 2026

# Problem set 2

The Posner cueing task is a classic experimental design in attention research. Use Python and PsychoPy to implement this task, as well as data analysis code, as follows.

Show a fixation point at the centre of a grey screen throughout the experiment. On each trial, show a sine wave grating for 500 ms at one of four randomly chosen locations relative to the fixation point: to the left, to the right, above, or below. Tilt the grating a few degrees to the left or right, with the direction randomly chosen on each trial. After the grating disappears, allow the subject to press 1 if they think it was tilted to the left, and 2 if they think it was tilted to the right. Play a high beep if their response was correct, and a low beep if it was incorrect.

A short time (500 ms) before the sine wave grating appears, show a cue for 100 ms. Make the cue a white outlined square, a bit larger than the sine wave grating. On 80% of the trials, randomly chosen, show a valid cue, meaning that the square appears at the location where the grating will appear later in the trial. Thus a valid cue helps the subject to move their attention to the correct location. On the other 20% of the trials, show an invalid cue, meaning that it appears at one of the other three locations, randomly chosen, where the grating will not appear.

To summarize, on each trial the sequence of events is: valid or invalid cue (100 ms), blank interval (500 ms), sine wave grating (500 ms), subject's response, and auditory feedback. Also include a brief pause between trials.

Program the experiment to run 100 trials. In addition to responses 1 and 2, also allow the subject to press 0 on each trial to quit the experiment.

There are several parameters in the experiment that you can choose for yourself, including the distance from fixation where the stimuli appear, the brightness of the cue, and the size, spatial frequency, and orientations of the grating.

On each trial, record the following information on a new line in a text file: trial number, valid or invalid cue, cue location, stimulus location, stimulus orientation (left or right), subject's response (left or right), whether the response was correct or incorrect, and response time. Save all these values as numbers, e.g., encode the valid/invalid condition as an integer 0 or 1, and encode the cue and stimulus locations as integers 1 to 4.

Write a data analysis script to load the text file, and find the proportion of correct responses and the mean response time, separately for valid and invalid cues. Show these values in a plot, and also print them in the console.

Email your solution to me (rfm@yorku.ca) in a single .zip or .tar file named with your last name in lowercase, e.g., murray.zip.

*Due December 4, 2026*